End-to-End Data Analyst Project

Telecom Network & Customer Experience Analytics

Business Context: A telecom company is facing increasing customer complaints, delayed payments, and unclear impact of network quality on customer retention. Leadership needs data-driven insights to prioritize network investments and reduce churn.

Your Role as a Data Analyst

You are responsible for performing an end-to-end analysis using Excel, SQL, Python, and Tableau. Your goal is to transform raw telecom data into meaningful business insights and executive dashboards.

Dataset Overview

1 customers – customer demographics and subscription type
2 network_usage – daily data and call usage by customers
3 network_issues – network complaints and resolution time
4 billing – monthly billing and payment behavior

Excel Tasks

1 Clean and standardize city names and dates
2 Create month and year columns
3 Identify high-usage customers using IF logic
4 Create pivot tables for city-wise issues

SQL Tasks

1 Create relational tables and primary/foreign keys
2 Write joins across all datasets
3 Calculate complaint rate per customer
4 Use window functions for monthly trend analysis

Python (Pandas) Tasks

1 Create customer risk score using functions and if-else
2 Use lambda for heavy usage flags
3 Perform exploratory analysis and visualizations

Tableau Dashboard Requirements

1 Executive KPI dashboard
2 City-wise network performance
3 Customer risk and churn indicators

Final Business Questions

1 Which cities require urgent network improvement?
2 Does poor network quality affect payments and churn?
3 Are 5G users experiencing fewer issues?